18/1/2025(FN)

Hall Ticket Number:

MR-21

Code: B314CB6S3

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, December 2024/January 2025

Computer Organization and Architecture

(Common to CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Draw the block diagram of Computer?	2	2	1	2
1. b)	List out the arithmetic micro-operations.	2	1	1	2
1. c)	Distinguish the differences between micro programmed Control and Hardwired control.	2	1	2	2
1. d)	What is meant by addressing modes and types of modes?	2	3	2	2
1. e)	Do the subtraction using 2s complements 1010-1010011.	2	3	3	2
1. f)	Draw the hardware implementation of fixed point addition and subtraction.	2	4	3	2
1. g)	Compare any 2 points for main memory and auxiliary memory.	2	1	4	2
1. h)	List out the 2 types of memory techniques and explain any one memory technique of mapping.	2	3	4	2
1. i)	Explain the principle characteristics of multiprocessors. List its advantages.	2	3	5	2
1. j)	Illustrate the classification of pipelining.	2	2	5	2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain role of different computer registers with respect to instruction Fetch cycle and execution. Give example.	5	1	1	1
2. b)	Elaborate the memory-reference instructions with an example.	5	4	1	1
	OR				
2. c)	Explain Arithmetic, logic and shift micro operations with example.	5	4	1	1
2. d)	With a neat diagram explain basic functional units of computers.	5	1	1	1
3. a)	With the diagram explain the basic organization of micro programmed control unit	5	1	2	2
3. b)	Registers R1 and R2 of a computer contain the decimal values 1200, 4600. What is the effective address of the memory instructions? Identify the addressing modes.The values in [R1]=2,[R2-1]=4 i. Load R5,[R1] ii.MOV R5,#3000 iii. Store [R2],R5 iv. Add R5,-[R2]	5	2	2	2
	OR				
3. c)	With the diagram explain the basic organization of address sequencing.	5	1	2	2
3. d)	What are the basic differences between a branch instruction, a call subroutine instruction and a program interrupt?	5	2	2	2
4. a)	List out the steps in non restoring division method to perform 8/ 3.	5	2	3	3
4. b)	Design the flowchart of fixed point addition and subtraction in computer arithmetics with an example.	5	4	3	3
	OR				

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4. c)	Perform booth multiplication on the following numbers - 13(multiplicand), +11 (multiplier).	5	1	3	3
4. d)	Design the flowchart of floating point addition and subtraction in computer arithmetics with an example.	5	2	3	3
5. a)	With a neat diagram explain the memory hierarchy in a process.	5	3	4	2
5. b)	What is cache memory? What is locality of reference? With neat Diagrams explain the cache memory mapping functions.	5	2	4	2
	OR				
5. c)	With the neat diagram explain the concepts of Direct Memory Access (DMA).	5	1	4	3
5. d)	With the neat diagram explain the block diagram of Input-Output interface.	5	2	4	3
6. a)	Explain 8x8 omega switching network for multistage switching networks to control processor-memory communication in a tightly coupled multiprocessor system.	5	1	5	2
6. b)	Draw the block diagram of Parallel Processing.	5	2	5	2
	OR				
6. c)	Explain in brief the serial arbitration (daily-chain) procedure.	5	3	5	2
6. d)	Explain the functionality of arithmetic pipeline with an example.	5	3	5	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome